

Hanna Teshager

Data Scientist/AI Systems Engineer

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[LinkedIn Profile](#) | [GitHub](#)

Skills

Data scientist/AI systems engineer with 4+ years of experience building production-grade NLP pipelines and multilingual intelligence systems for African digital ecosystems. Expert in deploying Llama 3 via Groq API, architecting serverless ETL pipelines on AWS Lambda, and developing full-stack ML platforms using Django and PostgreSQL. Proven track record of processing 9M+ posts/day with 92% F1-score across 5 languages, reducing manual analysis time by 60%, and training 50+ journalists in OSINT techniques. Combines research-grade NLP with scalable cloud infrastructure to deliver actionable disinformation intelligence.

Multilingual LLM Ops: Llama 3, Groq API, PEFT, Hugging Face Transformers, Zero-Shot Classification, Amharic/Swahili/French/Bambara/Yoruba NLP, RAG Architecture, Safety Guardrails

Real-Time Data Pipelines: AWS Lambda, EC2, Selenium, ETL, PostgreSQL, Streamlit Cloud, Python, FastAPI

AI-Powered Analytics: BERT Clustering, TF-IDF + DBSCAN, LDA/NMF Topic Modeling, NetworkX, Gephi, Plotly, Social Network Analysis

Full-Stack ML Deployment: Django, JavaScript, Docker, Model Serialization (skops), CI/CD, API Development, Telegram Bot API

Domain Expertise: Disinformation Detection, FIMI, Election Monitoring, Coordinated Inauthentic Behavior (CIB), OSINT

Experience

Code for Africa/Investigative Data Analyst / AI Systems Engineer
August 2021 - Present, Remote

Automated Disinformation Detection Pipeline:

- Architected serverless ETL pipeline using AWS Lambda and Selenium to process 9M+ social media posts daily, achieving 35% accuracy boost in detecting coordinated inauthentic behavior.
- Deployed BERT-based clustering (TF-IDF + DBSCAN) to identify narrative amplification networks, reducing manual analyst workload by 60% through automated intelligence summaries.
- Built LLM-powered weekly reporting system that auto-generates election monitoring briefs from top narrative clusters, highlighting originators and virality tiers.

Multilingual Trending Topic Detection System:

- Deployed Llama 3 via Groq API for real-time zero-shot classification across 5 African languages (Swahili, French, English, Bambara, Yoruba), achieving a 92% F1-score without fine-tuning.
- Developed Streamlit Cloud dashboard unifying 1M+ multilingual posts

into a single analytical interface with robust text normalization and cross-platform trend tracking.

- Enabled real-time narrative interrogation during Côte d'Ivoire 2025 elections, allowing analysts to prioritize high-impact disinformation leads within minutes.

FIMI Vulnerability Index Platform:

- Led end-to-end development of full-stack ML platform measuring national susceptibility to Foreign Information Manipulation using Django, PostgreSQL, and JavaScript.
- Implemented real-time scoring engine using PEFT-adapted transformer ensembles and geopolitical risk adjustment modules for contextual vulnerability assessment.
- Integrated Groq-powered AI chatbot for natural language narrative interrogation, enabling non-technical stakeholders to explore complex datasets conversationally.
- Automated data ingestion via AWS Lambda + EC2, ensuring consistent deployment and scalable processing across environments.

Capacity Building & Training:

- Designed curriculum and trained 50+ journalists across multiple African countries in OSINT techniques, data storytelling, and malign network mapping.
- Produced technical documentation and workshops adopted by partner organizations for regional disinformation monitoring programs.

Education

University of Gondar/Bachelor of Science in Information Science
2016 - JUNE 2018, GONDAR, ETHIOPIA

University of Gondar/Master of Science in Data Science and Analytics
2019 - JUNE 2020, GONDAR, ETHIOPIA

Research work

Cardiovascular Disease Risk Prediction using Machine Learning Techniques
[Springer](#), [ResearchGate](#)

Certification

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- Cisco Certified Network Associate (CCNA) routing and switching training program at the University of Gondar Cisco Academy Center, 2018.
 - Python for Data Science and AI Coursera Certified
 - Data Engineering Essentials Coursera Certified
 - Optimizing and Deploying LLM Systems Coursera Certified

Conferences

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- Data Science Africa (DSA) annual summer school and workshop, Addis Abeba, Ethiopia, June 2019
 - IndabaX Ethiopia Annual Conference, National Artificial Intelligence Forum, Wolayta Sodo, Ethiopia, May 2019
 - ICFJ: Investigating disinformation & Influence operations workshop, Dakar, Senegal, July 2022
 - ICFJ: Investigating religious extremism and mis/disinformation workshop, Kano, Nigeria, June 2023